

Biology, meaning the science of all life, is a late notion.

- Leon Kass

LOOKING BEYOND AN MBBS

Careers in Life Sciences

At last, career planning in India is waking up and taking notice of the life sciences. Newspapers are dedicating entire sections to the breakthroughs in the biological sphere — from advancements in stem cell technology to GM foods and oil-spill cleaning bacteria, this field is making waves like never before, with a wide variety of career options. So, if your teenager is passionate about biology, medical college is not the only career option to consider!



What is life science? What are the various career options?

Life science is defined as the branch of science that deals with living organisms - plants, animals and human beings. It encompasses several fields – botany, zoology, pharmacology, biochemistry, etc.

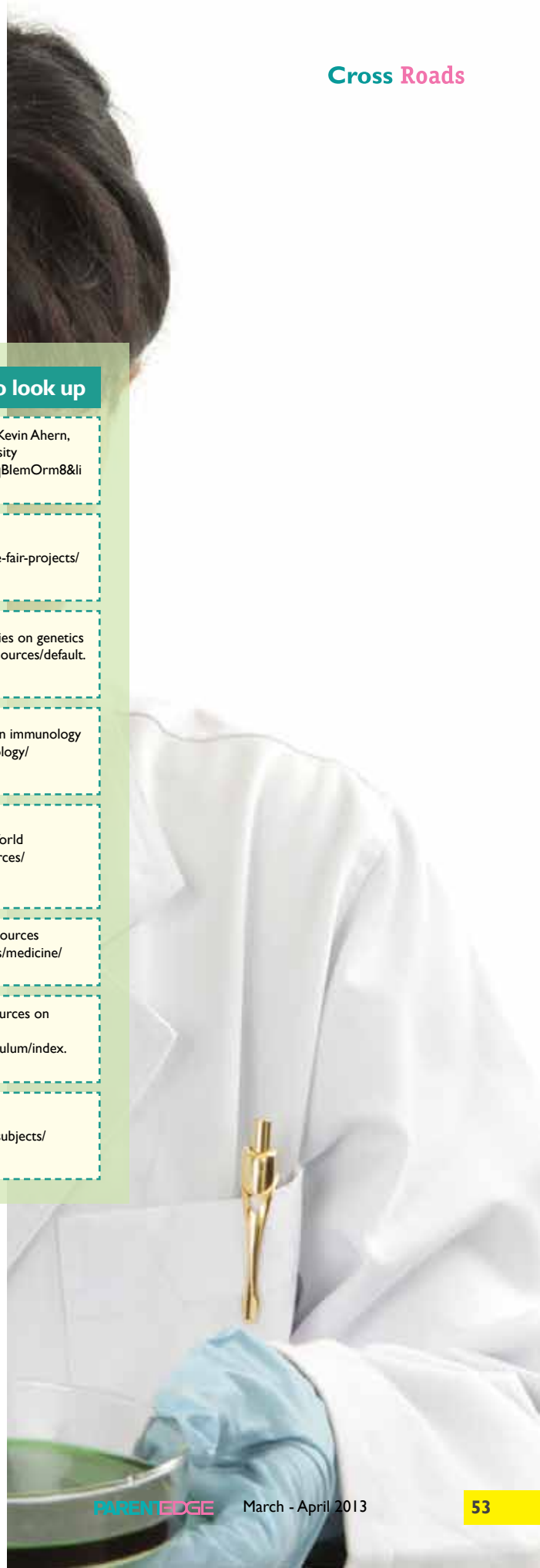
Branch	What it is	Where you work
Biochemistry	Study of the chemical composition and processes in living beings	In hospitals, pharmaceutical and food processing companies
Botany	Identification and classification of plants, the study of the structure of plants, the causes and cures of plant diseases	Botanical gardens, ecological consultancy agencies, food growers and biotechnology firms
Genetics	Study of the process of inheritance; investigation of how traits and disorders are inherited, and methods to alter or produce new traits	Health care, food processing, brewing, horticulture, beauty care, agriculture industries
Immunology	Study of how the human body protects itself against parasites, viruses and how it reacts to replacement organs	Pharmaceutical, biotechnology companies, diagnostic services
Microbiology	Study of microscopic organisms such as bacteria, viruses, algae, yeasts and moulds, and the impact of these on animals, plants, and the environment	Hospitals, pharmaceutical companies, food and beverage industries, water processing plants, bottling industries
Pharmacology	Development of new or improved drugs	Pharmaceutical and biotechnology companies
Toxicology	Planning and execution of investigations to determine the impact of toxic materials on living beings	Food product companies, cosmetics companies
Zoology	Study of the origin, behaviour, diseases, and life processes in animals	Zoos, non-profit organisations, museums and environmental consulting firms

A wide spectrum of organisations including healthcare, cosmetic and food-processing companies, NGOs, environment consultancy firms, and drug manufacturing companies hire specialists in life sciences. There are jobs available in areas such as management, administration, sales, legal support, writing and illustrating.

"It's a common misconception that much of the work in life sciences is research-based and limited to just laboratories," explains Dr. Raj Rajur, Chairman and CEO of CreaGen Biosciences, USA. "Research work, publications and new inventions eventually take you to executive positions. Also, once you have training in life sciences' research, you can even become an advisor or

Places to study	Interesting resources to look up
AIIMS, New Delhi (M. Sc., Ph. D.) JIPMER, Pondicherry (M. Sc.)	Introductory biochemistry lectures by Kevin Ahern, senior lecturer at Oregon State University http://www.youtube.com/watch?v=HHqBlemOrm8&list=PL69D27DEB1DF47B16&index=1
University of New Delhi (M. Sc., M. Phil., Ph. D.)	Fun botany projects to do http://www.factmonster.com/cig/science-fair-projects/liquids-do-seeds-grow-best.html
Jawaharlal Nehru University, Delhi (M. Sc.) Vellore Institute of Technology, Vellore (M. Sc. in biomedical genetics)	Worksheets, fun and educational activities on genetics http://www.cmri.org.au/Educational-Resources/default.aspx
National Institute of Immunology, New Delhi (Ph. D.)	Educational quizzes and problem sets on immunology http://www.biology.arizona.edu/immunology/immunology.html
Delhi University, New Delhi (M. Sc., Ph. D.) Doon Paramedical College, Dehradun (M. Sc.)	Projects and experiments at MicrobeWorld http://archives.microbeworld.org/resources/experiment.aspx
AIIMS, Delhi (M. Sc., Ph. D.) Amrita Institute of Medical Sciences, Kochi (M. Pharm. in pharmacology)	Science Fair Projects - Topics, Ideas, Resources http://www.juliantrubin.com/fairprojects/medicine/pharmacology.html
Jamia Hamdard, New Delhi (M. Sc., Ph. D.) University of Madras, Chennai (M.Sc., M. Phil., Ph. D.)	Activities, presentations and other resources on toxicology http://coep.pharmacy.arizona.edu/curriculum/index.html
Delhi University, Delhi (M. Sc., M. Phil., Ph. D.) Loyola College, Chennai (M. Sc.)	Useful links on zoology http://www.student-manual.com/study/subjects/zoology.htm

consultant to venture capital firms who fund new companies. If you have entrepreneurial inclinations, then you can establish your own company, as I did." Adds Sucharita Jayakar, Associate Scientific Manager at Biocon India, "Those with a business bent of mind can take up an MBA after a degree in any of the life sciences to get into pharma/biotech project management and strategy planning. The healthcare field is also rife with opportunities: for example, clinical research companies are always on the look-out for talented clinical trial managers."



Useful attributes for careers in life sciences

Analytical, communication and interpersonal skills are valuable skills that every aspirant should cultivate. "In my opinion, humility is the first attribute that one requires. It is a truly humbling experience to see the immense body of work and advances that have already taken place, and yet there is so much we do not understand! Therefore, a thirst for knowledge is required in order to keep oneself updated; in research, a determination to succeed is certainly a required characteristic," contributes Sucharita. "The most important attributes students need to have is interest in biological sciences and creative thinking. Computational skills and IT experience can also be helpful," says Dr. Rajur.

"You also need to have a desire to help people, high ethical standards, and the ability to work in multi- cultural, multi-ethnic and multi-disciplinary teams," adds Dr. Sheldon M. Schuster, President-Keck Graduate Institute of Applied Life Sciences, California, USA.

Educational path

The basic eligibility for a career in life sciences is a science background at the 10+2 level. There are many institutions in India which offer bachelor's degree courses in biotechnology, microbiology, zoology, botany and biochemistry. Many colleges also offer bachelor's degree programmes in life sciences. Following a bachelor's degree, one can opt for a master's in the fields mentioned above or in a variety of other specialisations such as immunology, pharmacology, genetics, toxicology, etc.

Some of the top colleges that offer bachelor's degree programmes are:

- Biotechnology – IIT Delhi, Kanpur, Kharagpur; AIIMS, New Delhi; Madurai Kamaraj University, Madurai.
- Life sciences – University of Delhi; JNU, Delhi
- Biology – IISc, Bangalore; AIIMS, Delhi
- Biochemistry – University of Delhi; Osmania University, Hyderabad
- Microbiology – University of Delhi; University of Pune

Resources

Books

- The Double Helix: A Personal Account of the Discovery of the Structure of DNA by James D. Watson
- A Primate's Memoir: A Neuroscientist's Unconventional Life among the Baboons by Robert M. Sapolsky
- The Cartoon Guide to Genetics by Larry Gonick and Mark Wheelis

Competitions:

- National Science Olympiad
  www.sofworld.org/nso
- Indian National Biology Olympiad
  www.ibo-info.org/countries/india/

Websites:

- Videos, slides, simulations
  www.hippocampus.org/
- Trivia quizzes on topics such as Cell and Molecular Biology, DNA and RNA, Evolution
  www.funtrivia.com/quizzes/sci__tech/biology.html
- A resource site with quizzes, worksheets and videos for biology enthusiasts
  www.biologycorner.com/
- Biology games and fun quizzes
  www.execulink.com/~ekimmel/games.htm



Interviews with Practitioners



Sucharita Jayakar is Associate Scientific Manager at Biocon India. She completed her M.S. in cell and molecular biology at the California State University at Hayward, USA.

What motivated you to take up cell biology after your under graduation?

After completing my under graduation in Nutrition and Dietetics and obtaining a Master's degree in French, I moved to the United States. During my stay there, I had time to reflect on what really interested and moved me, and I was drawn back to biology. I was amazed when I read papers from scientists who spent years of their life studying a protein folding process that takes place in a matter of a few seconds in our body, every day of our lives! I went in for higher education in cell biology and was fascinated by the layers of checks and balances and feedback loops in the working of the microscopic cell that controls life's processes.

Please describe the nature of your work. What are the career prospects for life sciences in India?

My work involves the formulation of biological molecules that are being developed as treatments for cancer and several other diseases. Formulation involves finding the right mix of ingredients and determining the right storage conditions so that the drug remains stable and active until it is given to the recipient.

This is a very exciting time to be in biological research in India, where opportunities exist to work with cutting-edge technology and collaborate with international resources. Avenues for learning are numerous, especially in the newly evolving fields of nanotechnology, proteomics and delivery devices.

What can a high-schooler do to prepare for a career in life sciences?

Read, get to know more. Today the resources available online are unmatched to anything in the past. A student must not miss any opportunity to participate in science fairs where the emphasis is on hands-on, practical applications of science. There are several companies that offer internships and traineeships for college students. These will help the student both in getting on-the-job training, and to evaluate where/how they fit in, in a company setting.

Do you know of any websites for high-schoolers to look up?

1. There is a vast collection of online interactives, web features, activities to explore science, from the Exploratorium Museum based in San Francisco, USA. (www.exploratorium.edu)
2. For learning concepts through a series of videos: www.khanacademy.org
3. Learning resource from Harvard University Life Sciences Outreach Program: http://outreach.mcb.harvard.edu/materials.htm



Dr. Raj Rajur is Chairman and CEO of CreaGen Biosciences, a drug discovery service-based company located in Boston, USA. Raj has authored more than 25 papers in peer-reviewed journals and holds three US patents.

Could you describe your own educational and career path? What motivated you to take up this stream?

I pursued medicinal chemistry and pharmaceutical sciences and obtained a Ph.D. in medicinal chemistry. I moved to the USA for postdoctoral studies at Boston College and Harvard Medical School. My motivation was to develop new medicines, especially for treating diseases like cancer and diabetes.

How closely is chemistry related to life sciences?

Chemistry is an integral part of all life science disciplines. Everything we use in our daily life has a component of chemistry in it. In fact, chemistry as a second major in a bachelor's degree course will be an asset for a career in life sciences.

What can a high-schooler do to prepare for a career in the life sciences?

High-schoolers can cultivate analytical thinking, creative problem-solving, and logic – these will be very helpful for a career in life sciences. It is also important to have a solid grasp of science and its relevance in daily lives. We at CreaGen provide a one week hands-on training for students interested in life science and biotechnology disciplines. Harvard summer schools also provide excellent opportunities. (www.summer.harvard.edu/programs/secondary-school)



Dr. Sheldon M. Schuster, President - Keck Graduate Institute of Applied Life Sciences, California, USA.

"A strong background in any science - Maths, Physics or Chemistry - will help you for a career in Life Sciences, which is inter-disciplinary in nature. High schoolers should focus on 'doing' so grab any opportunity to do hands-on research and choose a bachelor's degree course that offers opportunities for research. Read a lot books on science across disciplines, biographies of scientists to learn about science."

- Medha Nainegli, 20 years, Bangalore